Physics 1502Q

Tutorial 6.2
Midterm 1 Exam Review

There is no correlation between these practice problems and the actual exam problems.

Problem 1

A positive point charge Q is fixed on a very large horizontal frictionless tabletop. A second positive point charge q is released from rest near the stationary charge Q and is free to move. Which statement best describes the motion of q after it is released? **Justify**.

A. Its speed will be greatest just after it is released.

B. Its acceleration is zero just after it is released.

C. As it moves farther and farther from Q, its acceleration will increase.

D. As it moves farther and farther from Q, its acceleration will decrease.

Problem 2

A sphere with charge *+q* is placed a distance *d* from an uncharged metal sphere. Of the four figures shown, which figure best represents the resulting charge distribution on the metal sphere? **Justify.**

A. Figure A

B. Figure B

C. Figure C

D. Figure D

E. None of these

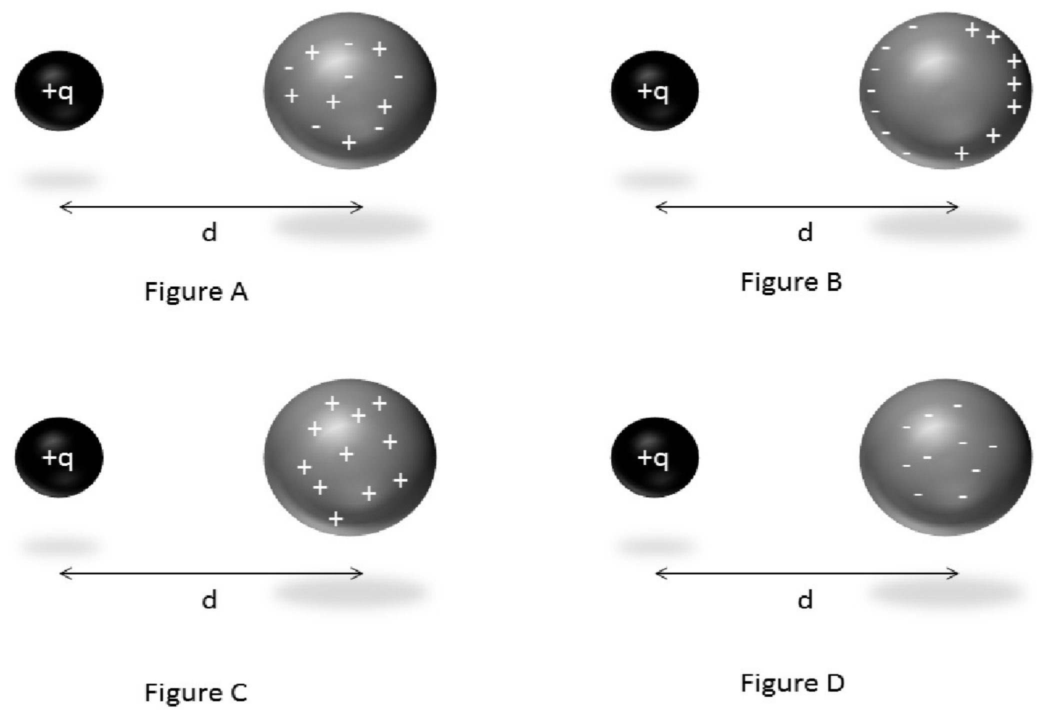


Figure A

Figure B

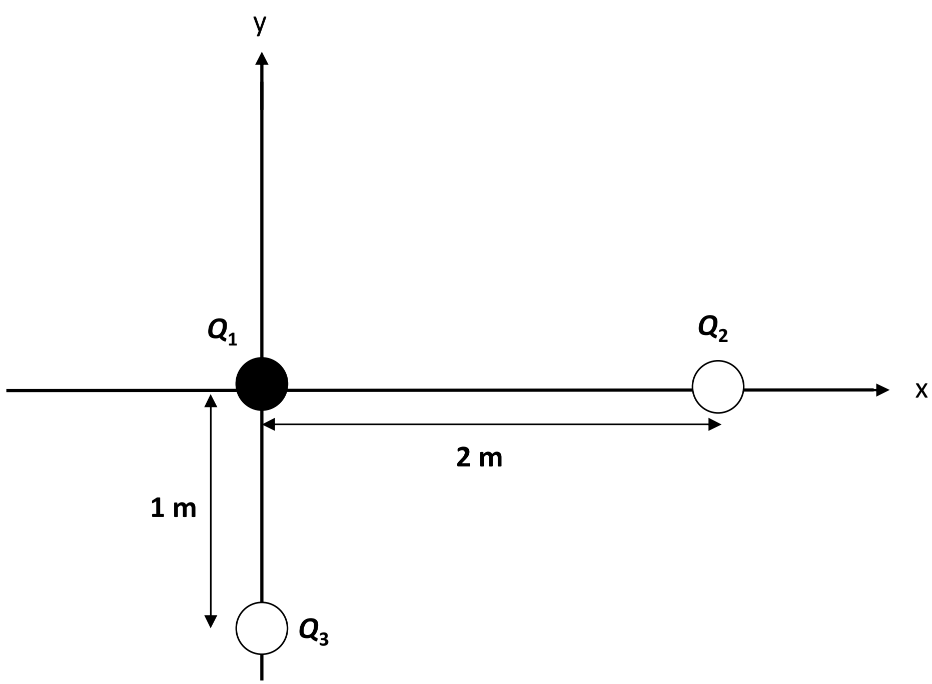
Figure C

Figure D

Problem 3

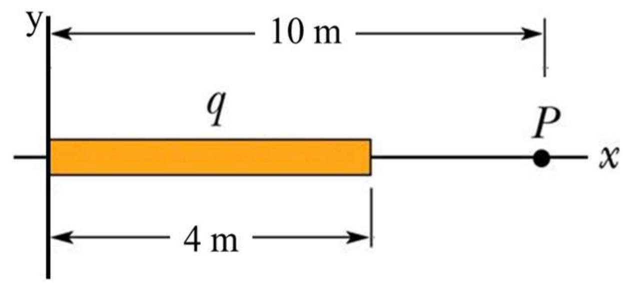
Three charges are arranged as shown in the diagram below. All three charges have a value of +1 μC. What is the net force on charge Q_2?

Q_1 = +1 μC
Q_2 = +1 μC
Q_3 = +1 μC



Problem 4

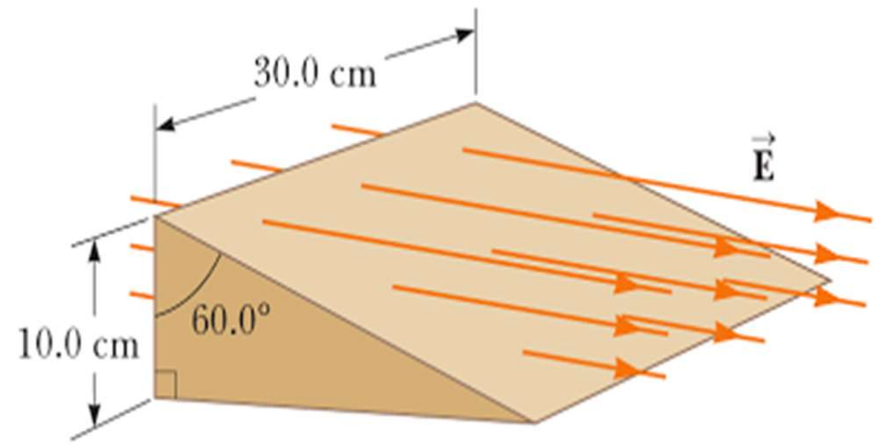
A thin rod has a total charge q distributed uniformly along the x-axis from $x = 0$ to $x = 4$ m. Set up the integral that gives the x-component of the electric field at point P at a distance of 10 m from the origin on the x-axis?



Problem 5

Consider a closed triangular box resting within a horizontal electric field of magnitude $E = 7.80 \times 10^4$ N/C as shown in the figure.

Calculate the electric flux through the vertical rectangular surface.



Problem 6

What is the sign of the flux through the slanted surface? <u>Justify.</u>

A. $\Phi_{slanted} > 0$
B. $\Phi_{slanted} < 0$
C. $\Phi_{slanted} = 0$

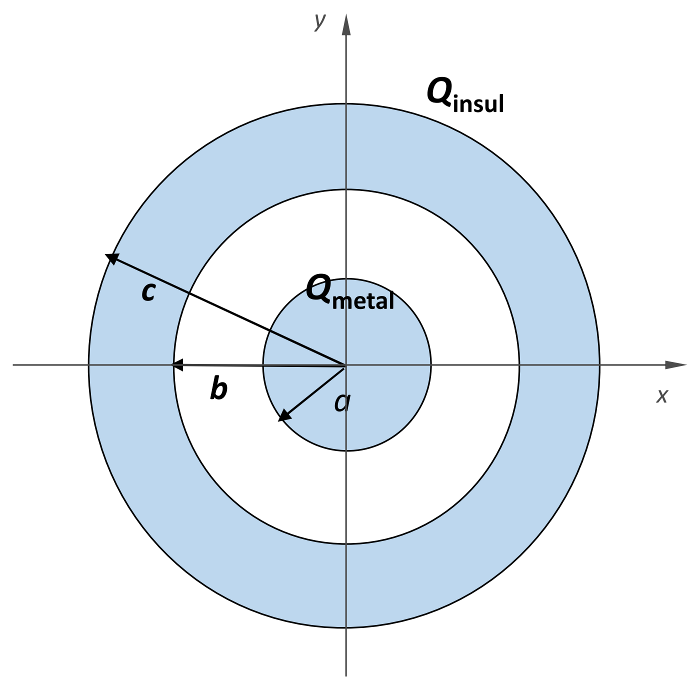
Problems 7, 8 & 9

A long metal (perfectly conducting) cylinder of radius a and length L is concentric with the z-axis. Surrounding it, and also concentric with the z axis, is a cylindrical shell made of insulating material; it is of the same length L, has inner radius b and outer radius c. (Both cylinders are so long compared with their radii that they may be considered of infinite length) A total negative charge Q_{metal} is placed on the inner metal cylinder, while a total positive charge Q_{insul} is uniformly distributed over the volume of the outer insulating cylinder. The values of all parameters are given in the figure.

7. Calculate the magnitude of the electric field E at a radius of 5 cm from the z-axis.

8. What is the surface charge density σ_{metal} on the outer surface of the metal cylinder?

9. Calculate the magnitude of the electric field E at a radius of 7 cm from the z-axis.



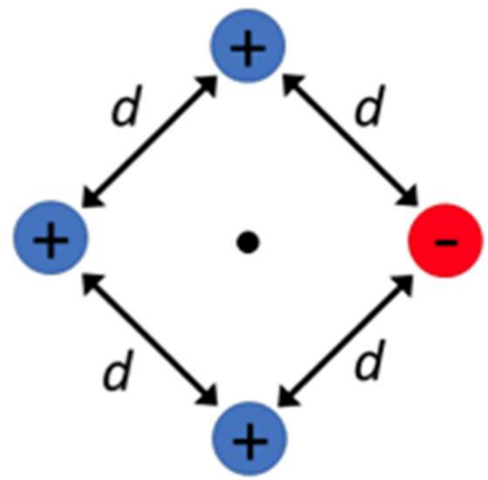
Both cylinders are of length L = 25 m (into the page)

a = 3 cm
b = 6 cm
c = 9 cm
Q_{metal} = –5 mC
Q_{insul} = +2 mC

Problem 10

Four point-charges are equally spaced by a distance $\boldsymbol{d = 8\ cm}$ at the corners of a square as shown in the figure. Three of the charges are positive with $\boldsymbol{q = 2.9\ \mu C}$, while one is negative with charge $\boldsymbol{q = -2.9\ \mu C}$.

(a) What is the total potential energy of the configuration?

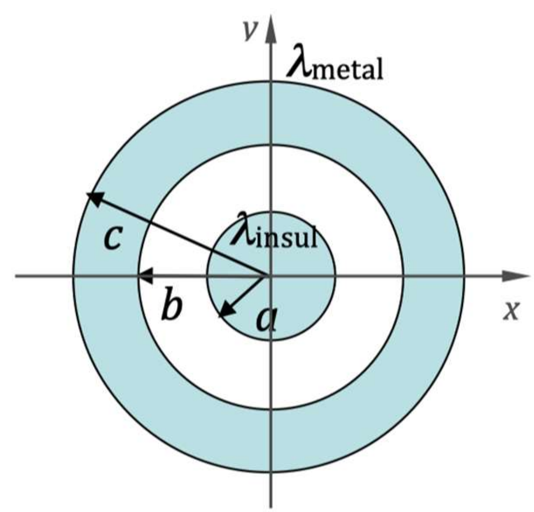


(b) What is the electric potential at center of the square?

Problem 11

A long cylinder of radius a and length L is made of insulating material and is coaxial with the z-axis. Surrounding it, and also coaxial with the z-axis, is a cylindrical shell made of metal; it is of the same length L, has inner radius b and outer radius c. (Both cylinders are so long compared with their radii that they may be considered of infinite length.) Line-charge densities of λ_{insul} and λ_{metal} are placed on the inner cylinder and outer shell respectively. The values of all parameters are given in the figure.

Find the potential difference $V_a - V_c$ between the surface of the insulating cylinder ($r = a$) and the outer surface of the metal shell ($r = c$).



Both cylinders are of length $L = 350$ m (into page)

$a = 5$ m
$b = 11$ m
$c = 17$ m
$\lambda_{\text{insul}} = -8 \cdot 10^{-9}$ C/m
$\lambda_{\text{metal}} = +5 \cdot 10^{-9}$ C/m